# Perimeter and Area

## Test 3

Date: _____ Duration: 1 hour Maximum marks: 30

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Use $\pi = \dfrac{22}{7}$. Draw a rough diagram wherever it helps. Show all your steps.

**Section A (1 mark each)**

*Questions 1 to 4 are multiple choice. Questions 5 and 6 are assertion–reason.*

**1.** A circular rangoli and a square rangoli each have a perimeter of 88 cm. Which statement is true?

(a) The circle has the larger area

(b) The square has the larger area

(c) The two areas are equal

(d) The areas cannot be compared without more information

**2.** A sector of a circle has radius 14 cm. The length of its arc is 22 cm. What is the area of the sector?

(a) 77 $\text{cm}^2$ (b) 154 $\text{cm}^2$ (c) 308 $\text{cm}^2$ (d) 616 $\text{cm}^2$

**3.** A triangle and a parallelogram stand on the same base and lie between the same parallel lines. The area of the parallelogram is 48 $\text{cm}^2$. What is the area of the triangle?

(a) 12 $\text{cm}^2$ (b) 24 $\text{cm}^2$ (c) 48 $\text{cm}^2$ (d) 96 $\text{cm}^2$

**4.** A protractor is a semicircle of radius 7 cm. What is the perimeter of the protractor?

(a) 22 cm (b) 29 cm (c) 36 cm (d) 44 cm

In Questions 5 and 6, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

**5.** Assertion (A): A circle with circumference 44 cm has an area of 154 $\text{cm}^2$.

Reason (R): The area of a circle $= \dfrac{1}{2} \times$ circumference $\times$ radius.

**6.** Assertion (A): A median of a triangle divides it into two triangles with equal perimeters.

Reason (R): A median of a triangle divides it into two triangles with equal areas.

**Section B (2 marks each)**

**7.** A circular cover for a well has a diameter of 14 cm. Ravi finds its area like this:

$$\text{Area} = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2.$$

Explain Ravi's mistake. Find the correct area of the cover.

**8.** The area of a circular park is 616 $\text{m}^2$. Find the length of wire needed to fence the park once.

**9.** A lawn sprinkler sprays water up to a distance of 21 m. It turns through an angle of 120° and back. Draw a rough diagram. Find the area of the lawn that gets water.

**Section C (3 marks each)**

**10.** A triangular plot of land has sides 26 m, 28 m and 30 m.

(a) Find the area of the plot using Heron's formula.

(b) The owner wants to divide the plot into two parts of equal area with one straight fence from a corner. Where should the fence meet the opposite side? Give a reason, and find the area of each part.

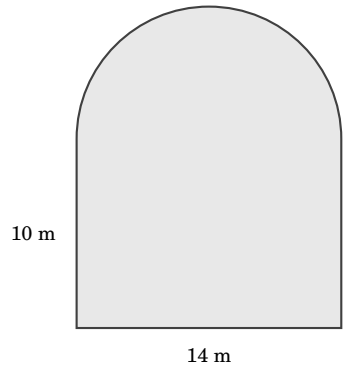
**11.** In a circle with centre $O$ and radius 14 cm, a chord $AB$ makes an angle of 90° at the centre.

(a) Draw a rough diagram. Mark $O$, $A$, $B$, the radius and the angle. Shade the minor segment.

(b) Find the area of the minor segment.

(c) Find the area of the major segment.

**12.** A school stage is a rectangle 14 m long and 10 m wide, with a semicircle on one of the 14 m sides, as shown.




(a) A rope is fixed along the whole edge of the stage. Find the length of the rope.

(b) Find the area of the stage floor.

**Section D (5 marks)**

**13.** A circular pond in a temple garden has a radius of 14 m. A path 7 m wide runs around the pond, outside it.

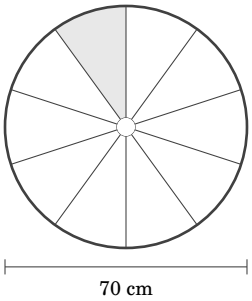
(a) Draw a rough diagram. Find the area of the path. [2]

(b) The path is tiled at ₹ 40 per $m^2$. Find the cost of tiling. [1]

(c) Meena says, "If the path were 14 m wide, its area would be double." Is Meena right? Show your working and explain. [2]

**Section E (case-based question, 4 marks)**

**14.** *Case study: the bicycle wheel.* Lakshmi rides her bicycle to school. Each wheel has a diameter of 70 cm. Each wheel has 10 spokes, equally spaced, that run from the centre to the rim.




(i) Find the distance the bicycle moves when the wheel turns once. [1]

(ii) Lakshmi's school is 1.1 km from her home. How many times does each wheel turn on her way to school? [1]

(iii) Find the area of the part of the wheel between two neighbouring spokes. [2]

*OR*

(iii) Find the perimeter of the part of the wheel between two neighbouring spokes. [2]